PALEOMAGNETISM OF THE HALIBURTON INTRUSIONS: SUPERIMPOSED MAGNETIZATIONS, METAMORPHISM, AND TECTONICS IN THE LATE PRECAMBRIAN

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Abstract. Three components of remanent magnetization have been isolated by af and thermal demagnetization of highly metamorphosed dioritic and gabbro-anorthositic rocks from Haliburton County, Ontario, in the Grenville structural province of the Canadian Precambrian Shield. The predominant A component, a normal magnetization, defines a paleomagnetic pole at 142.5°E, 35.6°S (α_{95} = 6.3°, k = 37.6), near southeast Australia, in agreement with published poles from widely separated areas in the Grenville Province. Paleopoles for the normal B and reversed C magnetizations fall much farther north, at 172.3°E, 24.5°N (α_{95} = 15.7°, k = 9.4) and 167.4°E, 3.0°S (α_{95} = 6.6°, k = 53.5), respectively. The C pole is close to secondary poles of other studies, but the B pole has not been detected elsewhere within the Grenville Province. It lies among poles for late Keweenawan rocks of the Lake Superior area. The A and B components are frequently superimposed in individual samples.

The three components have distinctive coercivity and blocking temperature spectra, which we interpret as recording mineralogy and physical conditions (temperature and duration of heating) at different stages of slow cooling following high-grade regional metamorphism of the Grenville Province. They constitute a record, from a single locality, of apparent polar wander between about 1200 and 950 m.y. The A magnetization was acquired at high temperatures, the B at much lower temperatures. Although both B and C magnetizations may have been set during subsidiary thermal pulses, a single-stage heating-cooling model of Grenville metamorphism is not ruled out. The proposed age sequence is A-C-B, a sequence most naturally explained by convergence of separate Grenville and North American plates. However, one-plate models which would incorporate the A-C-B sequence into a 'Grenville Loop' in the North American polar wander path are also possible.

Introduction

Since the early work of Balsley and Buddington [1957] demonstrating that the Adirondack gneisses are unstably magnetized, metamorphic rocks have been avoided by most paleomagnetists. Exceptions are studies of Precambrian rocks, which are invariably metamorphosed to some degree, but even here attention has centered on areas of exceptionally low-grade metamorphism. For example, almost half of the paleomagnetic studies of the North American Precambrian Shield (Figure 1) have used essentially unmetamorphosed rocks from the Keweenawan Basin, representing only a few percent of the area of the shield and a rather narrow age span (perhaps as narrow as 1140-1120 m.y. according to Silver and Green [1972]).

Park [1973] has made the interesting observation that amphibolite grade acidic gneisses from an area in the Churchill structural province are unstably magnetized while similarly metamorphosed mafic rocks from the same area behave stably. Evidently, metamorphic grade is not the sole criterion for paleomagnetic suitability. Within the Grenville structural province where the metamorphic grade is generally high (normally amphibolite to granulite), a variety of mafic intrusive rocks [Hargraves and Burt, 1967; Murthy, 1971; Palmer and Carmichael, 1973; Irving et al., 1972; Park and Irving, 1972] and a metasedimentary iron formation [McWilliams and Dunlop, 1975] have yielded reliable paleopoles, confidence limits being comparable to those obtained with unmetamorphosed Phanerozoic rocks (cf. Table 4 of this paper).

In all studies of metamorphic rocks one must, of course, be careful to distinguish between the age of formation of the rock and the age of magnetization. Within the Grenville Province, Wynne-Edwards [1972] has recognized structural trends associated with the 1735-m.y. Hudsonian orogeny, but K/Ar ages have usually been reset to 950 ± 150 m.y. and Rb/Sr ages to 1200 ± 150 m.y. by Grenville metamorphism. (The term Grenville orogeny is usually restricted to the resetting of K/Ar ages.) The Rb/Sr date of 1200 m.y. can at best be taken as an average because Rb/Sr and U/Pb dates suggest (A. J. Baer, unpublished compilation of ages, 1976) that the culmination of regional metamorphism was not contemporaneous throughout the province. Since radiometric ages have usually been reset by the

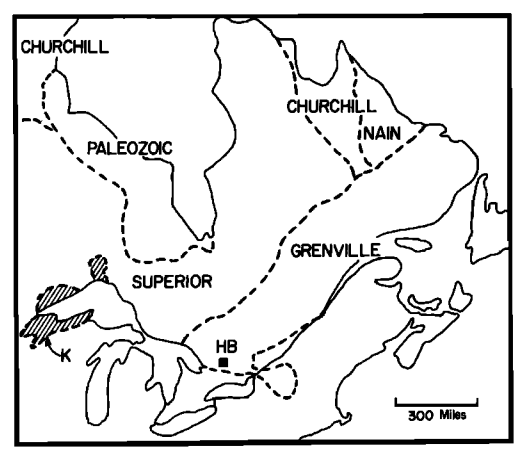


Fig. 1. Structural provinces of the Canadian Shield and location of the Haliburton intrusions (HB) and the Keweenawan igneous rocks (K).

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regional metamorphism, we might ask whether the same is true of primary magnetic remanences.

Buchan [1973], Buchan and Dunlop [1973], Fahrig et al. [1974], Irving et al. [1974a], and Ueno et al. [1975] have found that rocks of high metamorphic grade from the Grenville Province often carry two (occasionally three) components of NRM (natural remanent magnetization), each distinctly different in direction from the present earth's field (PEF). We shall refer to these as A and B remanences. Although the studies were all carried out on mafic intrusions (for the most part gabbro-anorthosite complexes), the occurrence of Grenville A and B components is not believed to be a feature peculiar to such rocks. Nor is it a local phenomenon, for the formations studied cover most of the 1900 km of exposed Grenville Province, from the Mealy Mountain anorthosite in the east to the Whitestone anorthosite and Haliburton intrusions in the west.

Since the A and B components are frequently superimposed in the same rock sample, they evidently represent spot readings of the earth's field at two different times. The blocking temperature spectra of A and B components [Irving et al., 1974a; Ueno et al., 1975; present paper] suggest that neither component predates the high-grade regional metamorphism nor postdates the close of the Grenville orogeny. In other words, superimposed Grenville remanences were probably all acquired between about 1200 m.y. and 950 m.y., perhaps during a single long cooling episode. However, more than one heating-cooling cycle may have been involved.

The directions of the A and B components shed some light on the long-standing 'Grenville problem' [Irving et al., 1974b; Baer et al., 1974]. The problem, stated briefly, is that paleopoles thought to be associated with Grenville metamorphism fall far to the south of both the North American polar wander path during the period 1200-1050 m.y. [e.g., Stewart and Irving, 1974] and all post-Grenville orogeny poles. A possible interpretation is that the Grenville Province was divergent from the rest of North America prior to the Grenville orogeny. However, the only geological evidence for a suture between the Grenville Province and the rest of the shield (W. R. A. Baragar's evidence in Baer et al. [1974], Brown et al. [1975]) would place the suture south of some localities where 'Grenville' paleopoles have been isolated [see Ueno et al. 1975; McWilliams and Dunlop, 1975] and therefore would not explain the discrepancy. A paleopoles from multicomponent studies fall near eastern Australia, like the typical Grenville poles of earlier studies. B paleopoles are often less well-defined, partly because they must frequently be separated from A components by vector subtraction, but they tend to lie at or north of the equator. They are thus in the same region as postorogeny Grenville poles and late Keweenawan poles. For this reason and on the basis of blocking temperature studies, Irving et al. [1974a] conclude that the high coercive force (hcf) A component in the Morin anorthosite is older than the medium coercive force (mcf) B component.

Although we shall defer a full discussion of the Grenville problem to a later section, we note in passing that the relative ages of A and B components are of crucial importance in tectonic interpretations. The divergence between Grenville poles and those from the other structural provinces of the shield implies either an undiscovered loop in the North American polar wander curve (and this is entirely possible, particularly since the resetting of NRM by great orogenies ensures a sporadic record) or the interesting possibility of independent 'Laurentian' [Stewart and Irving, 1974] and Grenvillian plates. Wilson [1966] has presented evidence for a Paleozoic proto-Atlantic ocean, whose closing left a suture parallel to but not coincident with the present-day Atlantic coastline. The Grenville Front, which is roughly parallel to both the Appalachians and the Atlantic seaboard, could on a two-plate Grenville model represent the suture (or be parallel to the suture) from a still earlier opening-closing cycle [e.g., Irving et al. 1974b]. But the two-plate model is possible only if the A poles are older than the B poles, for only then do these poles define a convergent juncture in the polar wander curve.

The purpose of the present paper is to give a full discussion of the A and B components of NRM in the Haliburton intrusions and to report evidence for a third C component not discussed by Buchan and Dunlop [1973]. We document in some detail our procedures for recognizing and separating superimposed components in the hope that these may be helpful to others working in metamorphic (not necessarily Precambrian) terrains. The coercive force and blocking temperature spectra of our A and B components appear to correspond closely to those reported by Irving et al. [1974a] for the Morin anorthosite and Ueno et al. [1975] for the Whitestone anorthosite, and so may possibly have general significance in multicomponent NRM studies. Furthermore, as we shall see in a later section, they are an essential clue to the relative ages of the components and hence indispensable in tectonic interpretations.

Geological Setting

The Haliburton intrusions (the Dudmon and Bark Lake diorites and the Glamorgan gabbro-anorthosite complex of Glamorgan and Monmouth townships, Haliburton County, Ontario) are located in the Hastings-Haliburton Highlands, a region of high-grade (upper amphibolite to granulite facies) metamorphism. The area lies in the western Grenville Province, a few miles north of the boundary of the exposed Canadian Shield and approximately 240 km south of the Grenville Front (see Figure 1).

Armstrong and Gittins [1968], who mapped the area, describe the country rocks of the Haliburton area as extensively and severely metamorphosed 'Older Granite Gneiss' complexes, of which the Glamorgan gneiss of Figure 2 is an example, and smaller petrologically distinct and younger gneiss complexes, which have been affected only by the late stages of metamorphism. They argue that two metamorphic episodes may be necessary to explain the metamorphic features evident in both series in view of the considerable length of time between their emplacement. On the other hand, MacIntyre et al. [1967] on

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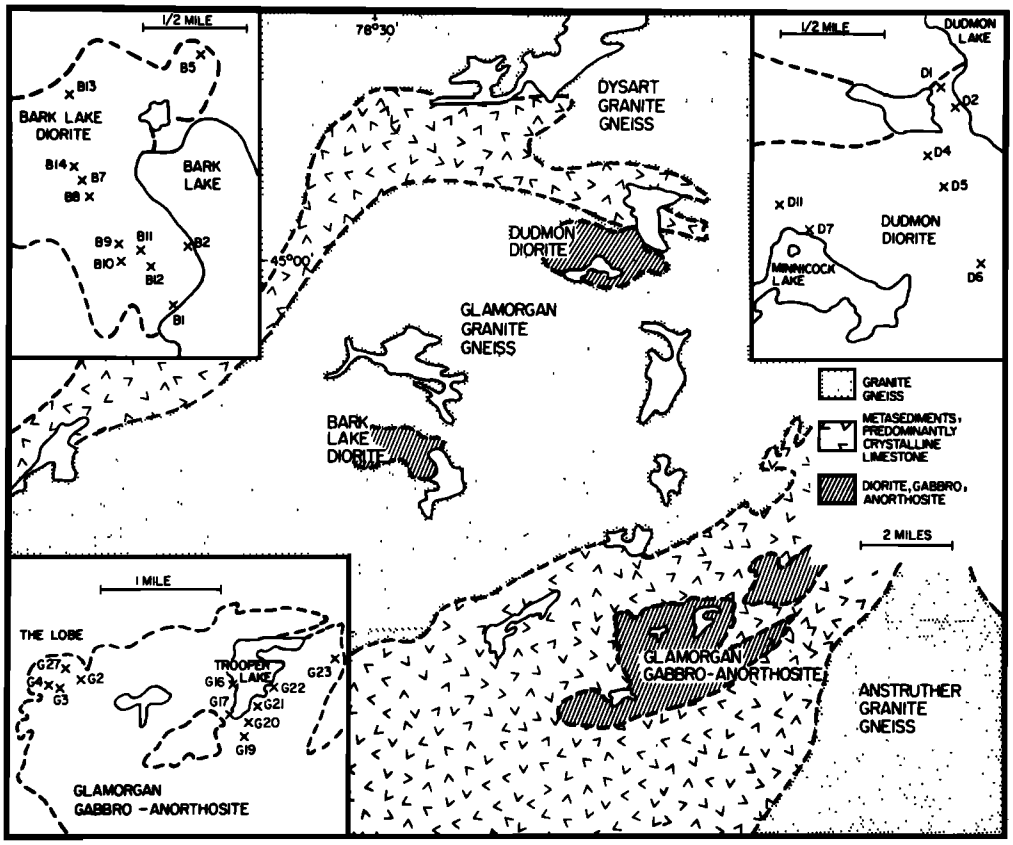


Fig. 2. Haliburton geology and sampling sites (inserts). Only sites mentioned in the text are shown.

the basis of radiometric dating (outlined below) favor a single prolonged period of regional metamorphism. Either sequence of events could account for the superimposed NRM components discussed in the introduction.

Geochronological data for the Hastings-Haliburton Highlands are incomplete. No radiometric ages are available for the Haliburton intrusions themselves. However, Silver and Lumbers [1965] assign U/Pb zircon ages of 1125 ± 25 m.y. and 1250 m.y. ± 25 m.y. to two groups of plutons in the Hastings-Haliburton Highlands. MacIntyre et al. [1967] report K/Ar mineral ages and summarize available K/Ar, U/Pb, and Rb/Sr dates for the area. They find two groups of ages: one at about 900 m.y. (based largely on K/Ar biotite ages), the other at about 1060 m.y. (based on U/Pb and Rb/Sr ages).

The Dudmon, Bark Lake, and Glamorgan intrusions are three of many mafic to ultramafic intrusions throughout the region. The intrusions exhibit distinct metamorphic features but primary igneous characteristics (e.g., flow line features and xenoliths) have been retained in some cases [Armstrong and Gittins, 1968; Downing, 1973]. Parts of the Glamorgan complex display such features, and it was for this reason that the Glamorgan body was first chosen for paleomagnetic study.

The Glamorgan complex, a sheetlike body intruded along an unconformity with the basement complex, is bounded on three sides by marble which has undergone tectonic flowage and folding and by syenite and paragneiss units conforming to the overall shape of the intrusion. Downing [1973] divides the complex into a number of petrologically and metamorphically distinct units. The 'lobe' a small projection on the northwest, consists of fresh gabbro retaining a number of primary igneous features, including fine-grained xenoliths in the coarse-grained matrix and igneous flow textures [Downing, 1973]. Armstrong and Gittins [1968] suggest that the lobe escaped deformational stresses because of the flow of the surrounding marble. Our best examples of A and B components in the Glamorgan complex came from the lobe, while the entire western projection of the body, consisting of metagabbro, gave weak unstable results. With the exception of an exposure of basement complex south of Trooper Lake and of nepheline gabbro north of the lake, the remainder of the body consists of southwest-northeast trending zones grading from gabbro in the north to gabbroic anorthosite in the south. The C component of NRM discussed in the following sections is largely confined to this region but is found in both the gabbroic and the anorthositic rocks.

The Dudmon and Bark Lake bodies are predominantly dioritic intrusions into the Glamorgan gneiss complex, with gradation into granodioritic compositions near the eastern margin of the Bark Lake intrusion. Both bodies appear to be compositionally and structurally simpler than the Glamorgan complex, although neither has been mapped in detail. It is possible that the Bark Lake and Glamorgan intrusions had a common parental magma [Downing, 1973].

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Sampling and Measurement Procedures

One hundred and thirty samples were collected at 51 sites during three successive field seasons. Of these, only about half (from the 29 sites shown in Figure 2) were paleomagnetically useful. The remainder of the collection proved too weak and/or unstable to measure. The B component of NRM has proved to be particularly elusive, both because of sampling difficulties (the B component is sporadic in its occurrence, even within a single site) and because the B direction was generally obtained by vector subtraction and is not as well defined as the A and C directions (cf. Table 2). However, while the definition of the mean B direction is not entirely satisfactory in a statistical sense, the existence of a distinct B component is not in doubt.

Orientations were measured with a Brunton compass. To test for deflection of the compass needle by the outcrop itself, a 'check' was made by sighting on the sample's orientation markings from a distance of at least 1 meter.

NRM measurements and statistical analyses were made by standard techniques. Several specimens of each sample were measured by using commercial spinner magnetometers in the course of af demagnetization in 8 to 12 steps to 1000 Oe. Demagnetization was carried out initially with an inductrol-controlled instrument incorporating random tumbling and later with a commercial instrument, demagnetizing along three perpendicular axes in succession. Because of the highly individual response of these metamorphosed samples, it was not possible to employ blanket cleaning techniques. Selected samples were af demagnetized along a single axis to 3000 Oe with the use of an integrated magnetometer system [West and Dunlop, 1971]. Additional samples were thermally demagnetized in 50° or 100°C steps to 650°C in a non-inductive furnace whose temperature was maintained within ±5°C for 1 hour before cooling in a null magnetic field.

Af Demagnetization Results

A component of NRM. The A remanence direction is based on 42 samples from 15 sites (6 Dudmon, 7 Bark Lake, and 2 Glamorgan sites) which reach a common stable end point in af's of 150 Oe or less (Figure 3). Large soft components, of presumed viscous origin, are removed below 150 Oe. In some cases a harder component is superimposed on the A remanence, as evidenced by a small increase of intensity above 150 Oe. An example is shown in Figure 3a. Vector subtraction of this harder component (by a method described in a later section) shows that it has a steep downwards inclination. The average direction is approximately reversed to the A direction and may represent a reversal of the A component. However, because of the uncertainties inherent in resolving a small component superimposed on a much larger (and nearly reversed) remanence, the separated component may equally well be a hard VRM (viscous remanent magnetization) acquired in the PEF. It is well-known [e.g., Biquand and Prévot, 1971; Dunlop, 1973] that VRM carried by very fine grains is relatively hard to af demagnetization. In any case, this PEF or reversed A component

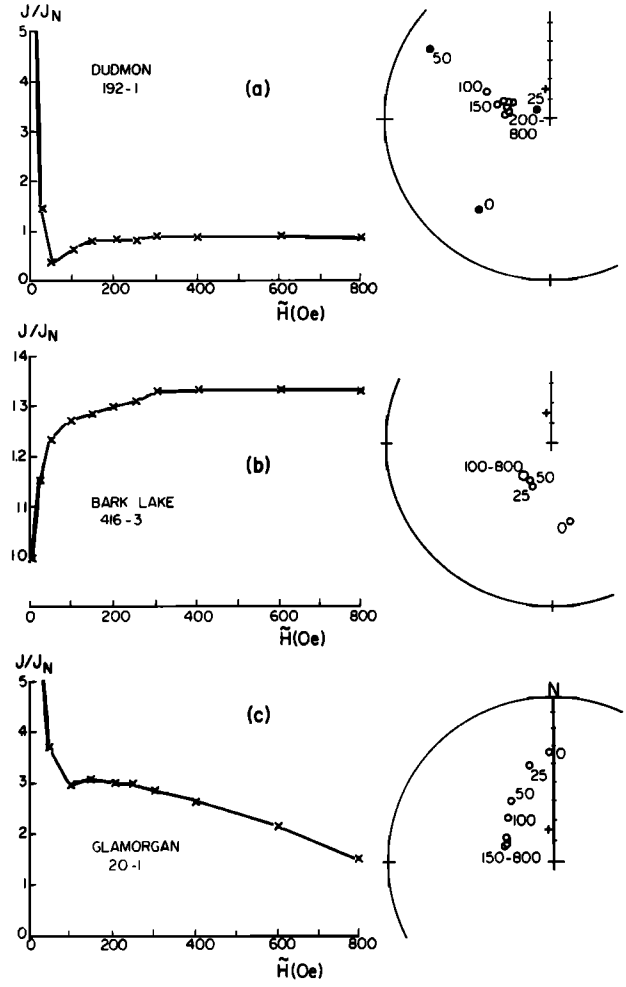


Fig. 3. The af demagnetization characteristics for typical A remanence samples. Directions are plotted on an equal-area projection with closed circles indicating positive inclinations and open circles indicating negative inclinations. The direction of the present earth's field (positive inclination) is shown as a cross.

scarcely affects the determination of the A direction, since it is at most a few percent of the A component and approximately reversed in direction.

In peak af's of 150 to 1000 Oe, Glamorgan (gabbroic) samples exhibit a slow steady decrease in intensity (Figure 3c), while Dudmon and Bark Lake (dioritic) samples have practically constant intensity. The hardness of Dudmon and Bark Lake samples is more striking still when demagnetization is carried out to 3000 Oe. It is apparent from Figure 4 that the A remanence is frequently unaffected by af's of 1500 Oe and may retain a significant fraction of its original intensity even at 3000 Oe. This behavior is reminiscent of that of the early Precambrian Modipe gabbro [Evans and McElhinny, 1966] and the middle Precambrian Michikamau anorthosite [Murthy et al., 1971], whose stable NRM's are carried largely by fine magnetite needles. It is also similar to the behavior of the hcf component of NRM in the Morin anorthosite, which likewise yields an A direction and is believed [Irving et al., 1974a] to be carried by hematite. The A remanence in

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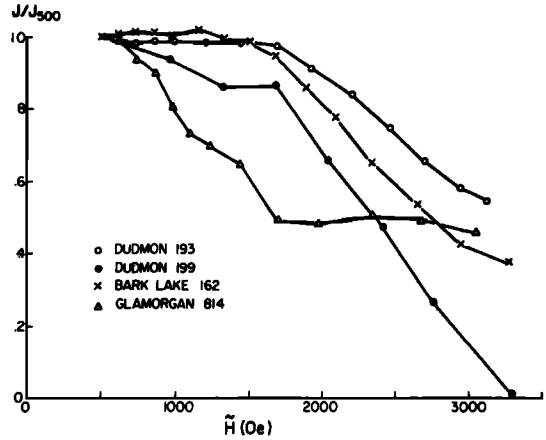


Fig. 4. A remanence af demagnetization characteristics for peak af's between 500 and 3000 Oe. All curves are normalized to the 500-Oe intensity.

our Bark Lake and Dudmon samples is carried largely by hematite, while the Glamorgan A remanence is carried by magnetite, as outlined in the section on thermal demagnetization.

B component of NRM. The B component of NRM is revealed most strikingly by remanence vectors, like those of Figures 5a-5c, which do not reach a stable end point in af's of 150 Oe but swing along great circle arcs of approximately 90° towards the A remanence direction between 150 and 1000 Oe. The direction of the B component which is superimposed on the A is not close to that of the PEF. As the B component is invariably much more intense than the A remanence on which it is superimposed and there is apparently no contamination by PEF components, the direction of the B component corresponds roughly to 100-Oe or 150-Oe directions in Figures 5a-5c. (A precise vector subtraction of the components is considered in a later section.) In addition to the nine samples exhibiting 'swinging' A + B directions, two samples (Figures 5d and 5e) maintain or reach a stable B direction on af demagnetization and show no sign of a coexisting A remanence. The B component is characterized by very shallow inclinations and declinations of 260° to 310°E.

The relative hardness of A and B components is illustrated by the demagnetization curves of Figure 6. The intensities are normalized to their 150 Oe values to ensure elimination of the bulk of any viscous components. The curves for the B component in swinging samples 302 and 139 were calculated on the assumption that the A component $\underline{J}^A$ is constant between 150 and 800 Oe (cf. Figure 4). This base line vector was computed by extrapolation (see section on vector subtraction) and then subtracted vectorially from the total remanence vector $\underline{J}$. It is evident that the vectorially subtracted B curves (302 and 139) agree closely with the directly measured B curves (149 and 857). The coercivity spectrum of the B component, although it is much softer than that of the A component, is moderately hard. In fact, a small percentage of the B component is retained at 1000 Oe. The B coercivity spectrum corresponds to that of the mcf component of the Morin anorthosite [Irving et al., 1974a] and perhaps to the northwest (NW) component of the Mealy Mountain anorthosite [Fahrig et al., 1974] as well. Both of these components, like the B component in our rocks, define paleopoles at or north of the equator.

C component of NRM. Both A and B components of NRM are normal magnetizations, defining north poles in the Pacific, although in the case of the A component the north paleopole is nearly rever-

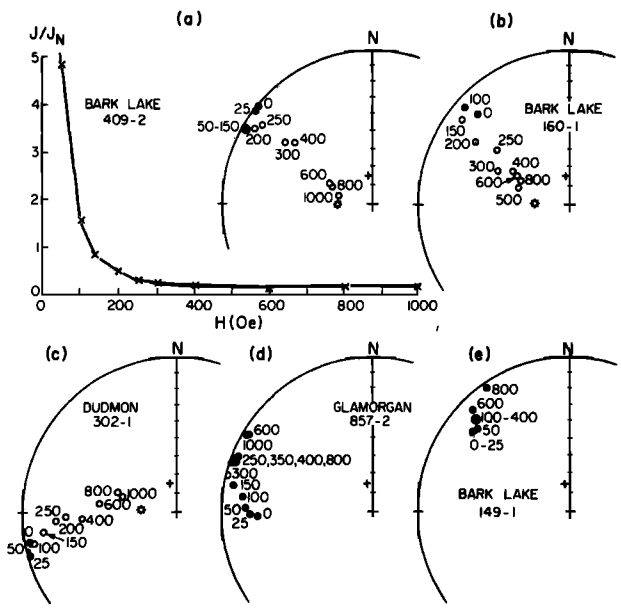


Fig. 5. The af demagnetization characteristics for typical B remanence samples. Symbols are the same as in Figure 3. The star marks the mean A direction. Three examples (a, b, c) exhibit a swing from the B towards the A direction in peak af's above 150 Oe. Two (d, e) maintain the B direction with no indication of a superimposed A component.

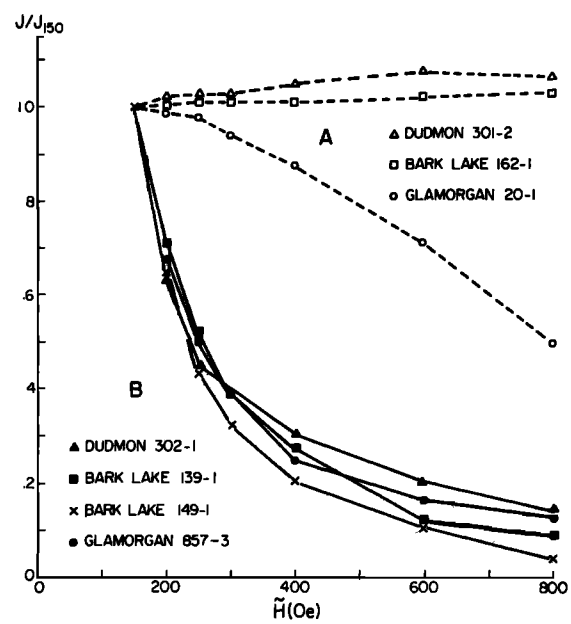


Fig. 6. Relative resistance to af demagnetization of A and B remanences above peak af's of 150 Oe. The method of calculating J/J_{150} for superimposed B samples is described in the text.

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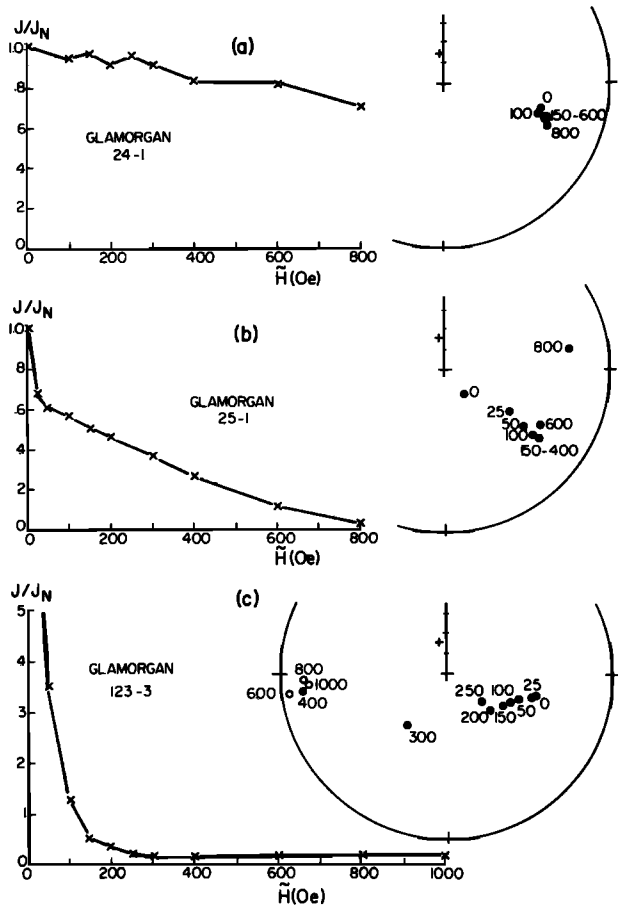


Fig. 7. The af demagnetization characteristics for typical C remanence samples. Two examples (a, b) exhibit only a stable C remanence. A third (c) is believed to contain superimposed B and C components. Symbols are the same as in Figure 3.

sed to the present north pole. A reversed magnetization distinctly different in direction from either A or B components is found at a number of sites in the Glamorgan intrusion. Often this C remanence has no superimposed A or B components (Figure 7a and 7b). However, at one site (G22), some samples swing from a B to a C direction, and at another site (G17) there is evidence of superposition of all three components. For example, specimen 3 of sample 123 from site G17 (Figure 7c) exhibits a steady swing from the C direction to approximately a B direction between 0 and 400 Oe, although two other specimens of the same sample maintain stable C directions over the same range. Whereas the A and B remanences usually occur in diorites or gabbros, the C remanence, although it is localized in occurrence, is encountered in a variety of rock types, metagabbro, anorthosite, and gabbroic anorthosite. The variable hardness of the C remanence (Figure 7) may reflect the varying mineralogies of these rocks.

Resolution of Superimposed Remanences

The general case of two superimposed remanences A and B is illustrated in Figure 8a. Since both $\underline{J}^A$ and $\underline{J}^B$ decrease in intensity between 150 and 800 Oe, it is clearly impossible to make a unique separation of the components. Even if the softer B component is totally erased, the only known quantity is the stable end point direction of the harder component $\underline{J}^A$, reached at high af's. (Note that the af's required to erase Grenville B components commonly exceed the limits of commercial af demagnetizers [e.g., Irving et al., 1974a; this paper]. In our case the direction assumed for $\underline{J}^A$ is the mean direction for samples exhibiting the A component only.) All one can say about the B direction is that it lies along the great circle defined by unit vectors in the directions of $\underline{J}_{150,400,800}$ and $\underline{J}^A$ and is no closer to the A direction than any of the difference vectors $\underline{J}_{150} - \underline{J}_{400}$, $\underline{J}_{150} - \underline{J}_{800}$, etc.

However, we can make a unique separation of A and B components if there is some range of af's over which $\underline{J}^A$ is constant in intensity. The process is illustrated in Figure 8b, where the A component is assumed to be constant from 150 to 800 Oe, as in the case of the Dudmon and Bark Lake rocks. The B direction is now defined by the vectors $\underline{J}_{150} - \underline{J}_{400}$ and $\underline{J}_{150} - \underline{J}_{800}$. Indeed, coincidence of these difference vectors was our test for the constancy of the A component between 150 and 800 Oe. The fields of 150, 400, and 800 Oe (or sometimes 1000 Oe instead) were chosen to satisfy the competing requirements that soft viscous components be eliminated, the A com-

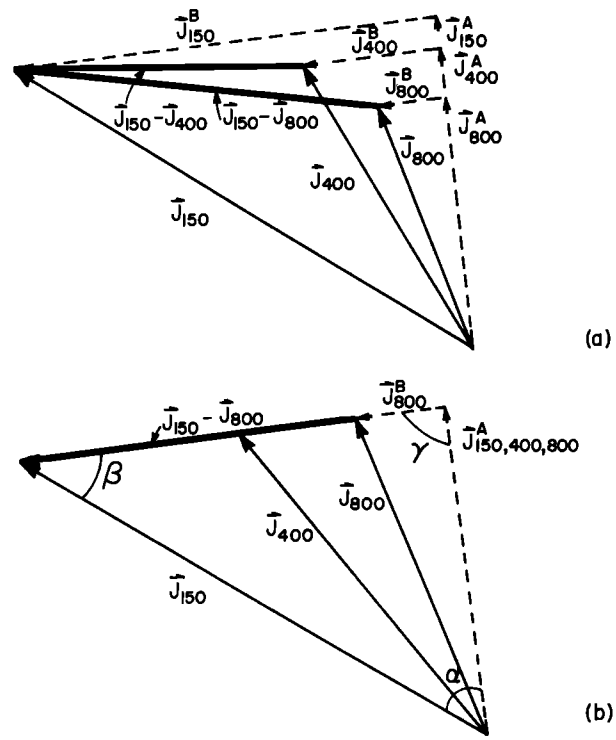


Fig. 8. Superimposed magnetizations. (a) represents the general case when two components are eliminated simultaneously. A unique resolution of the A and B directions is not possible. Subscripts denote peak alternating fields. Heavy arrows represent vector differences. (b) illustrates the procedure for isolating the direction and intensity of a superimposed B component when there exists some range (here 150-800 Oe) over which $\underline{J}^A$ is constant in intensity (see explanation in text).

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TABLE 1. Summary of A Remanence Directions and Paleopoles

Site	N	Mean Direction D	I	k	α_{95}, deg
B1	5	233.7°	-74.6°	54	12.7
B2*	2	267.2°	-57.2°		
B5(A)*	1	308.1°	-64.2°		
B8*	1	288.0°	-66.1°		
B9*	1	227.7°	-75.0°		
B12	5	212.4°	-71.7°	168	5.9
B14	3	269.5°	-62.3°	58	16.4
D1	3	253.7°	-67.6°	163	9.7
D2	3	279.8°	-69.9°	81	13.8
D4*	1	256.9°	-72.5°		
D5	4	165.6°	-83.3°	23	19.4
D6	4	266.0°	-72.0°	214	6.3
D11*	2	288.7°	-68.9°		
G3	3	344.1°	-66.2°	107	12.0
G4	4	293.2°	-67.4°	132	8.0
A mean, unit weight to sites	15	272.2°	-72.5°	37.6	6.3
A paleopole		142.5°E	35.6°S		
A mean, unit weight to samples	42	265.2°	-74.1°	27.4	4.3
A paleopole		141.7°E	40.0°S		

N is number of units (samples or sites) averaged, D and I are declination and inclination, k is Fisher's precision parameter, and α_{95} is the radius of the circle of 95% confidence about the mean direction.

Sites are the same as in Figure 2.

* Statistical parameters are not given for sites where N < 3.

ponent remain essentially constant, and the difference vectors be large enough not to be perturbed unduly by random and systematic measurement errors.

Coincidence between the difference vectors in samples exhibiting swinging A + B directions was excellent. This success can be attributed to the fact that the B component is more than an order of magnitude more intense than the A component in these samples. Separation of the small (a few percent of A) components which cause intensity increases above 150 Oe was less precise.

Once the direction of a superimposed component has been determined, its intensity can also be found. For example, the demagnetization curves for samples 302 and 139 in Figure 6 were constructed by determining the magnitudes of the 'base line' vector $\underline{J}^A$ from the relation $J^A = (\sin\beta/\sin\gamma)J_{150}$ (see Figure 8b) and subtracting $\underline{J}^A$ from $\underline{J}$ at each af. (The assumption here, that $\underline{J}^A$ remains constant up to the field, in excess of 1000 Oe, needed to erase $\underline{J}^B$, seems to hold well for Bark Lake and Dudmon samples.)

If the A direction of a particular sample is not given by the mean A direction for the site or formation but is displaced from it a small angle δ along a great circle perpendicular to that joining $\underline{J}^A$ and $\underline{J}^B$, α is virtually unchanged, and thus so is J^B. A displacement of δ along the great circle between $\underline{J}^A$ and $\underline{J}^B$ introduces an equal error in α, but since the present case, γ (the angle between $\underline{J}^A$ and $\underline{J}^B$) is nearly 90° and $J_{150}{}^B >> J_{150}{}^A$, α is also near 90°, and the error in calculating J^B is small.

Statistical Analysis and Paleopole Determination

A component of NRM. Paleomagnetic results for 42 samples (15 sites) exhibiting a stable A end point direction above 150 Oe are given in Table 1. The underlying A remanences $\underline{J}^A$ of swinging A + B samples are not included because they are less reliable. The site mean directions and their mean with its circle of 95% confidence are plotted in Figure 9a. The mean direction of magnetization giving unit weight to sites is D = 272.2°, I = -72.5° (k = 37.6, α_{95} = 6.3°), corresponding to a paleopole at 142.5°E, 35.6°S near southeastern Australia.

B component of NRM. The B component is widespread but very sporadic in its occurrence even

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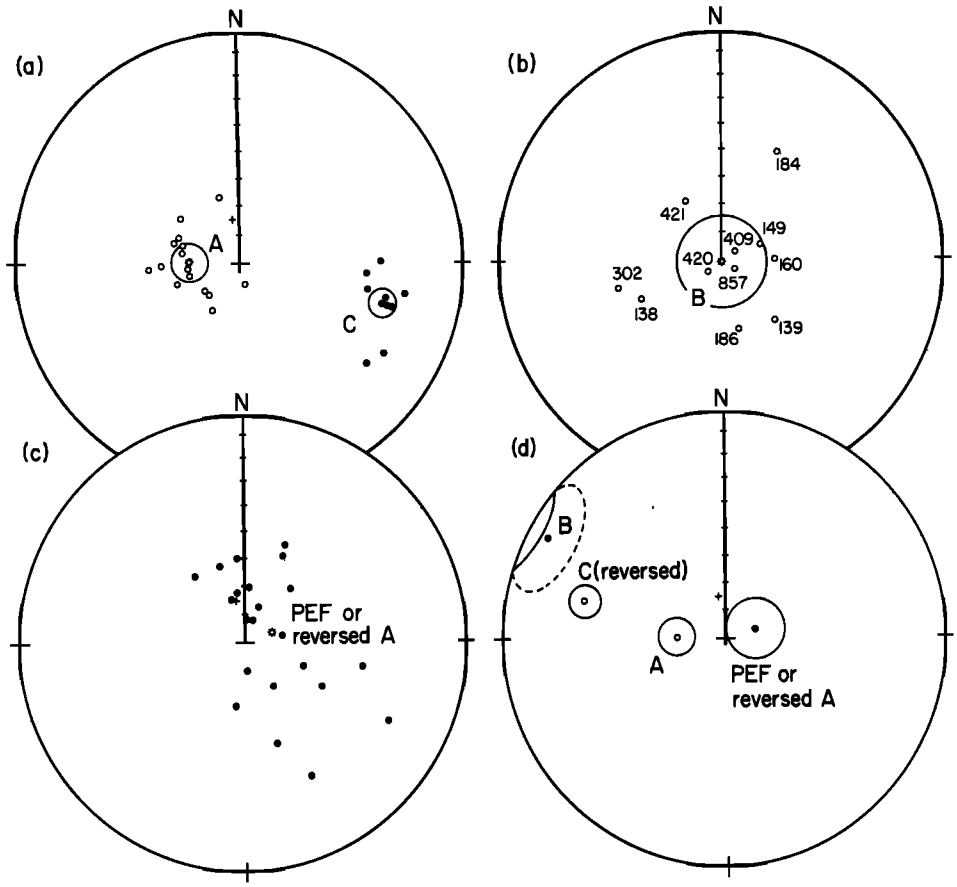


Fig. 9. Magnetization directions of the Haliburton intrusions. Symbols as in Figure 3. (a) represents A site mean directions (Table 1) and C sample mean directions (Table 3). In each case a mean (star) and circle of 95% confidence are drawn. (b) represents B sample mean directions (Table 2). Here the pole of the equal-area projection has been made to coincide with the mean B direction (D = 299.9°, I = 10.3°) in order to illustrate more clearly the statistical vector distribution. (c) represents PEF or reversed A sample mean directions and their mean. (d) represents A,B,C, and PEF or reversed A directions and their circles of 95% confidence. The C direction has been reversed to demonstrate its relationship with, but statistically significant deviation from, the A and B directions.

within a site, and the B directions are considerably more scattered than the A directions. The mean B direction in Table 2 is based on 11 samples from 8 sites (from all three intrusions). Two of these samples (Bark Lake 149 and Glamorgan 857) reach a stable end point B direction. Twenty-four individual specimens from the others yield a B component by vector subtraction.

Traces of the B component are found in at least two additional Glamorgan sites. Some specimens of sample 208 (site G2) exhibit swinging A + B behavior, but since other specimens of the same sample (and other samples from G2) are unstably magnetized in the PEF, the results from 208 were rejected. Samples from site G17 exhibit A + B + C components (see Figure 7c for an example of B + C behavior from this site), but the components cannot be uniquely resolved. Finally, two samples from Bark Lake site B13 and one from site B10 exhibit swings of about 90° between directions unrelated to either A or B directions. These samples were from small, poorly exposed outcrops and may be out of place.

Sample mean B directions and a circle of 95% confidence (unit weight to samples) are given in Figure 9b. (The pole of this equal-area projection has been made to coincide with the overall mean B direction to make clear the vector distribution.) The B mean direction giving unit weight to samples is D = 299.9°, I = 10.3° (k = 9.4, α_{95} = 15.7°) with a corresponding paleopole at 172.2°E, 24.5°N.

PEF or reversed A component. Table 2 also gives the results of vectorially subtracting the PEF or reversed A component. The vector subtraction was felt to be reasonably accurate only if the magnitude of the subtracted component was greater than 10% of the 150-Oe NRM intensity. Only results satisfying this criterion are plotted in Figure 9c. The scatter of directions is probably consistent with either a PEF or a reversed A component, although the mean falls near the reversed direction.

C component of NRM. The C component occurs in at least eight Glamorgan intrusion sites, the majority of them located in the eastern part of the body. Unfortunately, in some cases the C component is superimposed on an A or a B component, making a unique vector subtraction of the C component impossible. For this reason, only

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TABLE 2. Summary of Directions of B and PEF or Reversed A Superimposed Remanence Components and Corresponding Paleopoles

Sample	Site	N	Mean Direction D	I	k	α_{95}
149*	B2	3	315.0°	16.1°	379	12.9
138	B5	3	271.0°	-3.7°	44	18.7
139	B5	3	319.0°	-10.4°	254	7.8
409	B5	3	304.7°	13.8°	153	10.0
160	B7	3	318.6°	10.7°	33	21.9
420	B10	3	295.1°	6.6°	163	9.7
421	B11	3	285.2°	31.6°	201	8.7
302	D2	3	262.1°	-1.0°	31	22.7
184	G3	3	328.8°	47.7°	189	9.0
186	G3	3	306.3°	-13.7°	42	19.2
857*	G27	3	304.0°	8.0°	41	19.6
B mean, unit weight to samples		11	299.9°	10.3°	9.4	15.7
B paleopole			172.3°E	24.5°N		
PEF or reversed A mean, unit weight to samples		22	72.1°	78.4°	8.2	11.4

Symbols are the same as in Table 1 except that N represents number of samples or specimens averaged.

* Samples 149 and 857 exhibit B end point directions.

10 samples from six sites (Figure 9a) have been used in the statistical analysis of the C remanence (Table 3). These 10 samples reach a stable C end point direction upon af demagnetization.

Sites not considered in Table 3 include G17, where a unique resolution of superimposed A + B + C components is not possible (see, for example, Figure 7c), and G23, where some specimens exhibit a C direction while others are random or aligned with the PEF. In addition, B + C samples from site G22 were rejected.

Although only a rather small number of samples could be used for statistical purposes, thereby limiting the confidence one can place in the results, the mean NRM direction giving unit weight to samples (D = 105.4°, I = 34.7°) is well-defined (k = 53.5, α_{95} = 6.6°). The corresponding antipole lies at 167.4°E, 3.0°S, roughly on the great circle joining A and B paleopoles (see Figure 14).

Thermal Demagnetization Results

A component of NRM. A components of Dudmon and Bark Lake diorite samples have narrow blocking temperature (T_B) spectra. Blocking temperatures are usually above 600°C (Figure 10b) but occasionally are between 550° and 600°C (Figure 10a). In the former case, hematite evidently carries the A component. T_B's between 550° and 600°C are consistent with either magnetite or hematite, but several lines of evidence favor hematite as the sole carrier of Dudmon Bark Lake A components. First, all A samples, regardless of whether T_B's are above or below 600°C, have similar unusually hard coercivity spectra (cf. Figure 4). Although magnetite needles can, in principle, exhibit coercivities as high as 3000 Oe, it is more likely here that the high coercivities are in all cases attributable to hematite. Second, when samples with 550°-600°C T_B's and those with T_B's > 600°C are found at the same site, they have similar NRM intensities, saturation magnetizations, and other magnetic parameters. Finally, where T_B's above and below 600°C are present in a single sample, there is no evidence of overlapping T_B spectra due to two distinct phases.

Thermal demagnetization of a Glamorgan gabbro A sample is shown in Figure 10c. The directional data indicate that the decrease in intensity below 500°C is due, in part at least, to the elimination of a viscous component. Between 550° and 600°C there is a rapid intensity decrease and loss of directional stability. The af coercivity spectrum (Figure 3) is substantially softer than the spectrum of Bark Lake and Dudmon A samples with 500°-600°C T_B's, so that magnetite is the likely carrier of the A component in the Glamorgan rocks.

B component of NRM. Thermal demagnetization results for three samples exhibiting B components are shown in Figure 11. Two samples (Figure 11a, b) have stable B end point directions, while the third (Figure 11c) exhibits a directional swing towards the A direction. It is clear that the B

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TABLE 3. Summary of C Remanence Directions and Paleopoles

Sample	Site	N	Mean Direction D	I	k	α_{95}
19	G4	3	94.1°	43.2°	425	5.9
25	G16	3	127.0°	29.9°	1779	2.9
127*	G16	2	121.6°	26.8°		
110	G19	3	88.9°	38.0°	243	7.9
106	G20	3	101.1°	42.4°	64	15.6
109	G20	3	105.5°	34.1°	232	8.1
24	G21	3	105.2°	34.8°	542	4.0
113	G21	3	102.0°	34.4°	437	4.4
423	G22	3	100.2°	27.3°	161	9.7
424	G22	4	105.3°	31.4°	231	6.0
C mean, unit weight to samples		10	105.4°	34.7°	53.5	6.6
C paleopole			167.4°E	3.0°S		

Symbols are the same as in Table 2.
* Statistical parameters not given for N < 3.

component has a much broader spectrum of T_B's than the A remanence. A distinctive feature is the concentration of blocking temperatures below 400° or 450°C. In many cases, even the lowest

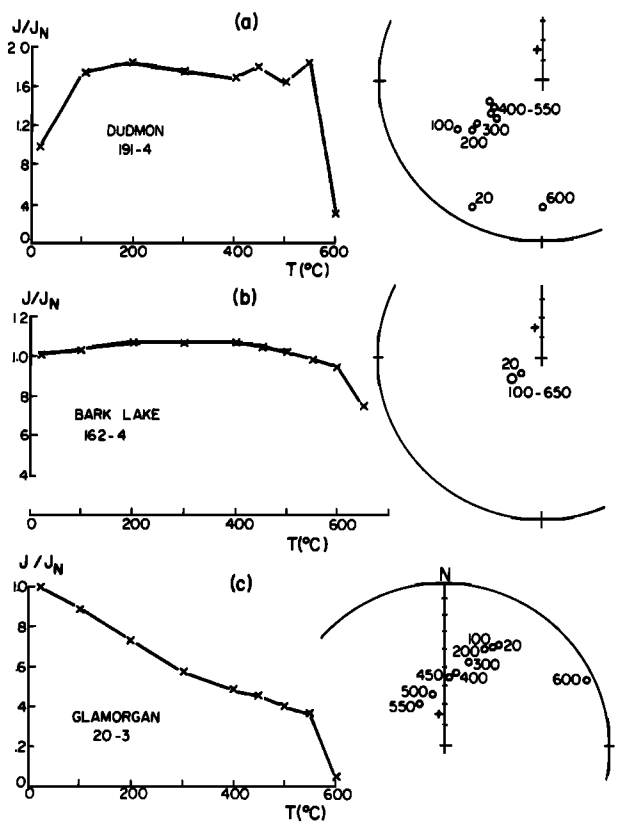


Fig. 10. A remanence thermal demagnetization characteristics. Symbols are the same as in Figure 3.

T_B's appear to be associated with the B direction (Figures 11b and 11c) rather than with a viscous component. A 2-yr storage test confirms the lack of viscous magnetization. However, despite the pronounced drop in intensity near 400°C, the directional stability of the B component is maintained to temperatures in excess of 600°C. We will refer to the phase carrying the bulk of the B remanence and having T_B's below 450°C as the B_1 phase and the smaller high-T_B phase as the B_2 phase.

C component of NRM. C remanence samples exhibit a variety of T_B spectra. Examples are shown in Figure 12. At an anorthosite site (Figure 12a) the C remanence is erased between 300° and 400°C, subsequent directions being random. In a metagabbro (Figure 12b), on the other hand, the direction and intensity stabilize between 400° and 550°C. Disappearance of the C direction at 600°C and above suggests magnetite as the dominant carrier of NRM.

Origin of NRM and Metamorphic History

The superposition of two (and occasionally three) NRM components with overlapping T_B spectra in the Haliburton intrusions records three episodes of magnetization. These episodes involve the effects of temperature, time, and chemical alteration, either separately or in combination, and are intimately related to the metamorphic history of the area. Deciphering the origin of the A, B, and C components of NRM is the key both to tectonic interpretations, in that it establishes the relative ages of A, B, and C paleopoles, and to an understanding of the Grenville metamorphism, e.g. whether or not a multistage heating-cooling model is required.

First, let us note that the absolute ages of all the NRM components must fall within a well-

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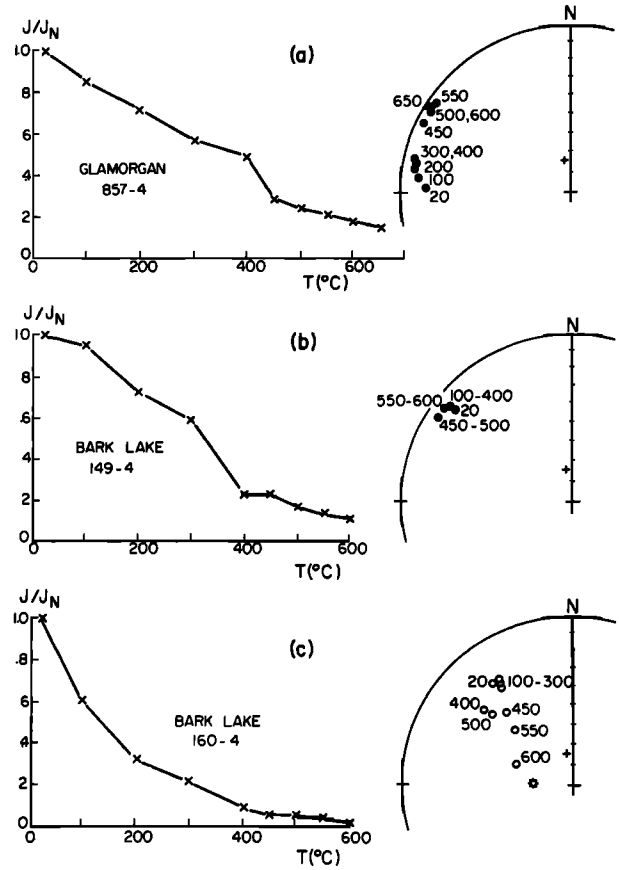


Fig. 11. B remanence thermal demagnetization characteristics. Symbols are the same as in Figure 3. The star marks the mean A direction. Two examples (a, b) reach stable B end point directions. The third (c) swings from the B towards the A direction.

defined range. The regional grade of metamorphism in the Hastings-Haliburton Highlands is upper amphibolite to granulite, implying heating to 600°-700°C. Although some of the primary NRM, with blocking temperatures above 600°C, could in principle survive such a heating and 'see through' the orogeny, this does not seem to have occurred. The A, B, and C components (Figures 10-12) either lose their directional coherence above 600°C or exhibit the same direction above as below 600°C. There is no sign of a directionally distinct primary NRM having only >600°C blocking temperatures. On the other hand, regional K/Ar dates of about 950 m.y. show that there has been no reheating above 200°C or so [Moorbath, 1967] since the close of the Grenville orogeny. Since all three directional components have T_B's in excess of 200°C and since all three have originated at least in part as thermal remanent magnetizations (see below), none can postdate the Grenville orogeny. For these reasons the A, B, and C components are likely bracketed by the high-grade regional metamorphism (∿ 1200 m.y.) and the close of the Grenville orogeny (∿ 950 m.y.). Comparable arguments have been used by Irving et al. [1974b] and Ueno et al. [1975] to set similar limits on the ages of multicomponent remanences in the Morin and Whitestone anorthosites, respectively.

Within this interval the relative ages of the components must be inferred by comparison of their detailed blocking temperature and coercivity spectra. The deductive process is not without its ambiguities, since higher blocking temperatures do not necessarily imply older magnetizations. For example, secondary hematite produced by oxidation below 500°C may carry a CRM (chemical remanent magnetization) whose blocking temperatures are higher than those of an older magnetite TRM. The model that we propose below is therefore not unique, but it is the simplest sequence of events that accounts for all the observations.

From the present evidence it is difficult to deduce the age of the C component relative to the other components. The C remanence does not exhibit a single characteristic T_B spectrum (probably because of the variety of rock types that carry the C remanence), and in the few cases where a C component is superimposed on A or B components it is impossible to make a clean separation of the coercivity and blocking temperature spectra into their component parts.

However, a clean separation can be made in swinging A + B samples, taking advantage of the characteristic A spectrum, shown in simplified form in Figure 13a. Assuming a similar concentration of A blocking temperatures above 550°C in swinging A + B samples, we derive the simplified T_B spectrum of Figure 13b, in which the only overlap is between the A and part of the B_2 spectrum.

Preliminary studies of polished sections of stable A samples indicate the presence of three main magnetic phases.

1. The first phase consists of large hemoilmen-

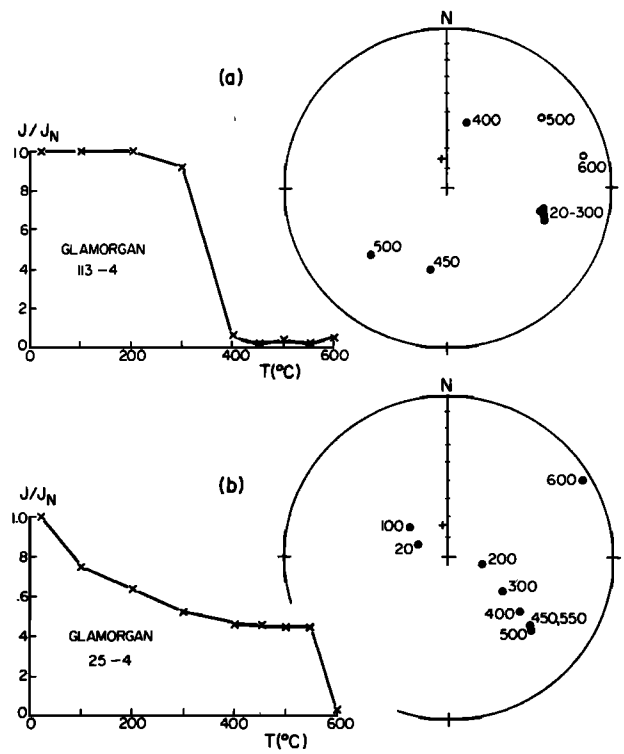


Fig. 12. C remanence thermal demagnetization characteristics. Symbols are the same as in Figure 3.

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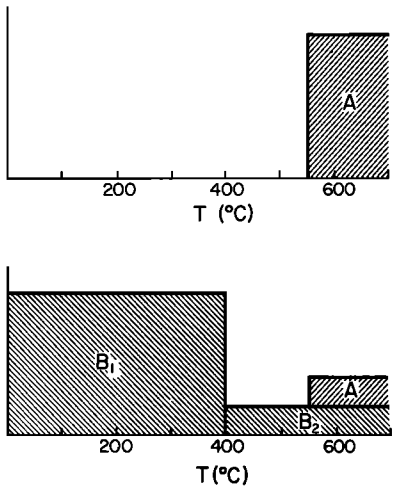


Fig. 13. Simplified blocking temperature spectra for (a) samples with only an A component and (b) samples with superimposed A and B components. Viscous components are not considered.

ite grains (100-500 μm) with patches and crystallographically oriented lamellas of exsolved hematite in an ilmenite host. These grains are the dominant oxides in Dudmon and Bark Lake samples, where they are undoubtedly the main carrier of the NRM. They have not been identified in the Glamorgan gabbro, in which homogeneous ilmenite is the dominant oxide, reinforcing our conclusion (previous section) that the Glamorgan A component is carried by magnetite.

2. A second phase consists of large, homogeneous, steel grey titanomagnetite grains, subhedral to anhedral, and predominantly 200-600 μm in size. Some grains show border granulation and minor martitization of some grains. These grains form up to 30% of the oxides in most rocks. Some samples contain a few percent of smaller (5-40 μm) homogeneous titanomagnetite grains.

3. A third phase consists of numerous fine (~1-5 μm) opaque rods and platelets in the silicates. Most are pyrite and hemoilmenite; none have been identified as magnetite.

Oxide occurrences similar to the phase 1 and 2 assemblage above are described by Irving et al. [1974a] for the Morin anorthosite 'primary' or hcf component samples (Mr_1 in Table 4) and by Ueno et al. [1975] for the Whitestone anorthosite WW samples (Ws_1 in Table 4). In both studies the A magnetization represented by these components is attributed entirely to hematite, in which case the A component could be either a TRM older than B component or a CRM, possibly younger than the B component. The fact that in our rocks both the magnetite and hematite can carry an A component, an observation made also by Hargraves and Burt [1967] and McWilliams and Dunlop [1975], rules out the second possibility, since there is no plausible mechanism for generating magnetite and hematite simultaneously at low temperatures. We conclude that the A component is a TRM (thermal remanent magnetization) acquired at high temperature.

The predominance of blocking temperatures below 400°C in B samples (Figures 11 and 13b) is unusual. It suggests that the B_1 remanence is carried by titanomagnetites with variable titanium content and was acquired as a viscous partial TRM (VPTRM) during very slow cooling. Using a characteristic time for the cooling process of 10 m.y., the maximum T_B of 450°C for the B_1 phase in laboratory thermal demagnetization corresponds to [Pullaiah et al.,1975] about 350°C during the original cooling. Irving et al. [1974a] note a similar concentration of low blocking temperatures associated with the mcf or 'secondary' (B) component of the Morin anorthosite (Mr_2 in Table 4), which they likewise interpret as indicating a late stage, low-temperature origin for the B remanence.

Overlap of the A and B_2 blocking temperature spectra (Figure 13b) indicates that only one of these components was acquired as a TRM at high temperatures. The B_2 component is therefore likely a CRM carried by hematite. Since B_1 and B_2 directions are the same, the hematite was presumably produced below 350°C and magnetized at the same time as the B_1 phase.

The simplest model of Grenville metamorphism consistent with the rock magnetic evidence for the A and B components is a single heating event followed by a slow cooling. Alternatively, the B component may have been set as a TRM(or VPTRM) by a mild reheating (the 950 m.y. Grenville orogeny) during the late stages of the cooling process. No reheating is possible after the acquisition of the B component, since the B direction is stable at blocking temperatures of 100°C or less. Therefore unless the C remanence is a CRM acquired essentially at room temperature, the B component is probably younger than the C as well as the A component.

The C component has blocking temperatures intermediate between the mean T_B's for the A and B_1 components, suggesting that the C component was acquired at intermediate temperatures during the slow cooling period which followed regional metamorphism. C blocking temperatures sometimes slightly overlap the highest T_B's (300°-450°C) of the B_1 component and the lowest T_B's (500°-600°C) of the A component. C components tend to occur in different rock types than A and B components and are probably carried by different mineral phases. Pullaiah et al. [1975] show that magnetite and hematite with identical T_B's on a laboratory time scale acquire TRM (or VPTRM)at distinctly different times during slow cooling over millions of years. Alternatively, one or more of the overlapping components may be a thermochemical remanence [Kellog et al. 1970], in which mineralogical changes continue below the blocking temperature. Whatever the exact nature of the C component, the observations are consistent with a single-stage cooling model of Grenville metamorphism, although multiple heating events or thermal pulses superimposed on the main cooling process are not ruled out.

Apparent Polar Wander and Tectonic Models

Irving et al. [1974b] have recently reviewed paleomagnetic results from the Grenville Province and the various tectonic models that could account for divergent Grenville poles. We shall therefore confine ourselves to evaluating the

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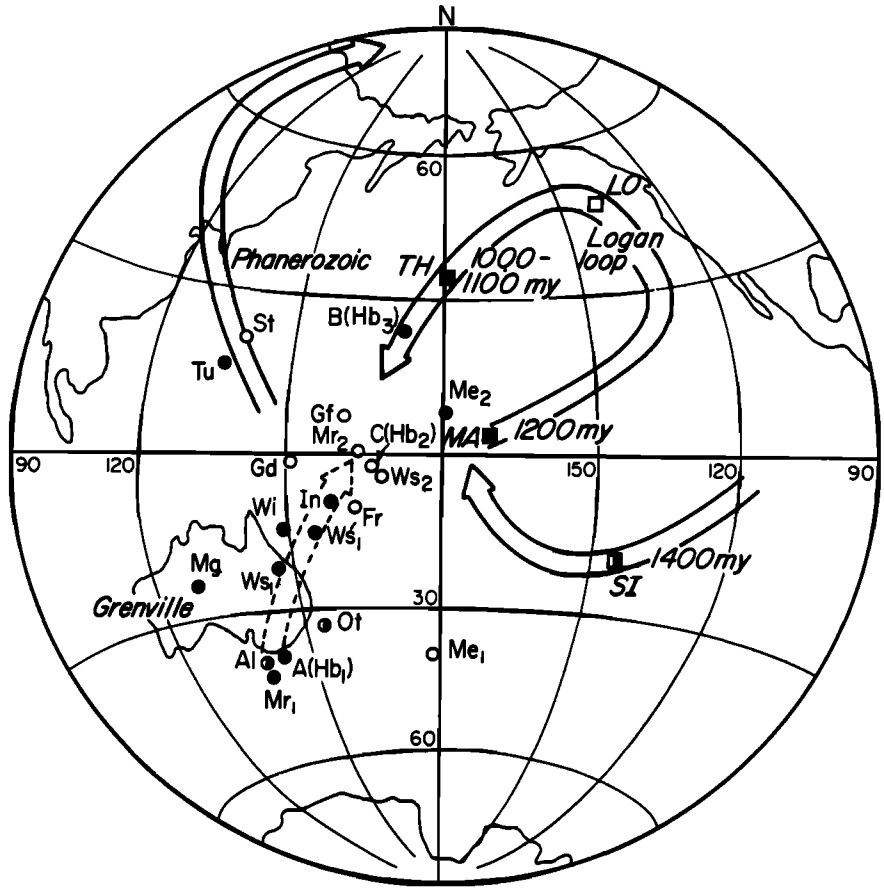


Fig. 14. Grenville paleopoles and the polar wander path for North America. Circles represent Grenville paleopoles and are labelled as in Table 4. Squares represent tie points used by Robertson and Fahrig [1971] and Robertson [1973] to define the late Precambrian polar wander path for North America including the 'Great Logan Loop'. LO is the Logan sills; MA, the Mackenzie dikes; SI, the Sibley sediments; and TH, the Thunder Bay dikes. Closed symbols represent normal poles, open symbols reversed poles.

models in the light of the two main conclusions of the last section: (1) A, B,and C paleopoles all postdate the peak of high-grade regional metamorphism, and (2) the probable time sequence (from older to younger) of the Haliburton poles is A-C-B.

On the basis of magnetic properties, Grenville poles (Figure 14 and Table 4) fall naturally into two groups. Group A, near eastern Australia, incorporates A component results from multicomponent studies (our A or Hb_1, Me_1, Mr_1, Ws_1, Ws_1') and poles from heavily metamorphosed units (Al, In, Mg, Ot, Wi). Group B, near the equator, incorporates B component results in multicomponent studies (our C or Hb_2, Me_2, Mr_2, Ws_2) and poles which are believed to postdate the peak of regional metamorphism from single-component Grenville units (Fr, Gd, Gf). A poles are normal (Hb_1, In, Mg, Mr_1, Wi, Ws_1, Ws_1') or mixed (Al, Ot) with the exception of Me_1, which lies far from other A poles and may not belong to group A. B poles are uniformly reversed with the exception of Me_2 and our B (or Hb_3) pole. Hb_3 and Me_2 lie well north of the main B grouping, and, in view of their location and normal polarity, may in fact be associated with normally magnetized late Keweenawan volcanics of the Lake Superior region. Finally, poles St and Tu diverge from all other groups. Tu is from the lightly metamorphosed Tudor gabbro, recently dated by Hayatsu and Palmer [1975] at 670 m.y., and St is from the Steel Mountain anorthosite of Newfoundland, which has been dated at 481 m.y. [Murthy and Rao, 1975]. Both these poles likely postdate other Grenville poles.

The many models that have been proposed to explain group A Grenville poles fall in two general classes. Two-plate models [Irving et al. 1972; Palmer and Carmichael, 1973; Buchan and Dunlop, 1973; Stewart and Irving, 1974; etc.] hypothesize that A poles record the divergence of the Grenville Province from the rest of North America ('Interior Laurentia'). B poles date from the time of convergence of the Grenville and Interior Laurentian polar wander paths and hence record continental convergence. These models would be especially attractive if they could explain Grenville metamorphism itself as a by-product of the continental collision. One-plate models [Roy and Fahrig, 1973; Fahrig et al., 1974; Stewart and Irving, 1974; McWilliams and Dunlop, 1975], on the other hand, make no attempt to explain the orogeny and would simply incorporate Grenville paleopoles into the North American polar wander path.

The major evidence favoring two-plate models

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TABLE 4. Summary of Grenville Paleomagnetic Poles and Fisher Statistics

Symbol	Rock Unit	Reference	No. of Sites	k	α_{95}, deg	Paleopole	Polarity
Aℓ	Allard Lake anorthosite	Hargraves and Burt [1967]	4	88	10	39°S, 140°E	N and R
Gd	Grenville diabase dykes	Murthy [1971]	11	14	11	3°S. 151°E	R
Ot	Ottawa basic intrusions	Irving et al. [1972]	13	25	8	32°S, 155°E	N and R
Fr	Frontenac Axis dykes	Park and Irving [1972]	6	85	7	12°S, 162°E	R
Wi	Wilberforce pyroxenite	Palmer and Carmichael [1973]	4	∿100	6	14.5°S, 148°E	N
Tu	Tudor gabbro	Palmer and Carmichael [1973]	10	5 to >100	8.5	17°N, 137°E	N
Gf	Grenville Front anorthosites	Palmer and Carmichael [1973]	4	∿100	6	8°N. 161°E	R
	Haliburton intrusions	Buchan and Dunlop [1973]	A,* 10	64	6	39°S, 144°E	N
			B,* (13)†	(8)	(16)	(21°N, 171°E)	N
		Buchan [1973]	C,* (5)**	(55)	(10.5)	(1.5°S, 165°E)	R
Me_1	Mealy Mountain anorthosite	Fahrig et al. [1974]	E,* 14	17	9	38°S, 178°E	R
Me_2			NW,* 11	12	12	8.5°N, 181°E	N
Mr_1	Morin anorthosite	Irving et al. [1974a]	P,* 24	32	5	42°S, 139°E	N
Mr_2			S,* 13	17	10	0°, 164°E	R
Mg	Magnetawan metasediments	McWilliams and Dunlop [1975]	23	51	4	24°S, 130°E	N
Ws_1	Whitestone intrusions	Ueno et al. [1975]	W,* 5	96	8	16°S, 156°E	N
Ws_1'			Z,* 4	223	6	22°S, 146°E	N
Ws_2			Y,* 8	45	8	5°S, 168°E	R
In	Indian Head anorthosite	Murthy and Rao [1975]	6	30	12	9.5°S, 158.5°E	N
St	Steel Mountain anorthosite	Murthy and Rao [1975]	10	22	12	22.5°N, 138°E	R
A or Hb_1	Haliburton intrusions	present paper	A, 15	38	6	36°S, 142.5°E	N
B or Hb_3			B, 11**	9	16	24.5°N, 172.3°E	N
C or Hb_2			C, 10**	54	7	3°S, 167°E	R

* Component symbols used in original papers.
† Number of specimens averaged.
** Number of samples averaged.

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is the apparent uniqueness of Grenville A poles. Rb/Sr ages of 1200 m.y. believed [Wynne-Edwards, 1972] to date the culmination of regional metamorphism and hence close to the age of Grenville A remanences do not differ greatly from Rb/Sr ages for the Mackenzie dikes of the Superior and Churchill provinces. Yet the corresponding paleopoles diverge from each other by approximately 50°. In fact, no Precambrian paleopoles of any age fall in or near group A in Figure 14 except those from the Grenville Province.

A second point favoring two-plate models is the proposed age sequence (A-C-B) for the Haliburton paleopoles. The sequence is consistent with the closing of a proto-Atlantic Ocean between the Grenville and Interior Laurentia, collision between the two plates occurring at the time of the Haliburton B pole. The age sequence is less naturally explained by one-plate models, most of which propose a Grenville loop rather than a unidirectional A-C-B sequence.

Third, the grouping of Grenville A and B paleopoles according to polarity is reminiscent of long periods of single polarity noted in the North American data by Robertson and Fahrig [1971] and Robertson [1973]. Proceeding back in time from the proposed polar juncture, we have in Interior Laurentia [Robertson and Fahrig, 1971] the normal Thunder Bay dikes pole, the reversed Logan sills pole, the normal Mackenzie dikes poles, and the mixed polarity Sibley sediments pole (see Figure 14). Perhaps a corresponding sequence in the Grenville Province is the normal Haliburton B or Hb_3 pole, reversed group B poles, normal group A poles, and finally the mixed polarity poles Al and Ot, the most southerly and easterly of group A. The suggested broad correspondence between these polarity sequences is tentative and speculative. Nevertheless, if correct, it is an extremely valuable aid in assigning ages to Grenville remanences which are difficult to date by conventional radiometric methods because of the uncertainty in the temperatures (hence, for slow cooling, in the times) at which isotopic clocks and magnetizations are reset [Pullaiah et al., 1975].

One obvious difficulty with two-plate models is the lack of evidence of a suture zone at or near the Grenville Front, except possibly just to the north and east of the Haliburton area [Brown et al., 1975]. Elsewhere preorogenic geological units cross the front [e.g., Wynne-Edwards, 1972] and can be recognized as far as 100 km into the Grenville Province.

Furthermore, high-grade regional metamorphism of the Grenville Province cannot be explained simply as the result of collision between the two plates. We have concluded that regional metamorphism had reached its peak before the A remanence was acquired, hence, in any two-plate model, at a time when the Grenville and Interior Laurentia were still far apart. The subsequent collision of these plates was sufficient only to rest low and moderate blocking-temperature remanences (B's and C's) along with K/Ar mineral ages. However, it can be argued (H. C. Palmer and R. Van der Voo, personal communications, 1975) that plutonism and metamorphism associated with subduction of the leading edge of the Laurentian plate beneath the Grenville could be responsible for regional metamorphism, the subsequent continental collision bringing the activity to a halt and perhaps causing a final moderate reheating (the 950 m.y. Grenville orogeny).

Thus although paleomagnetic results from the Haliburton intrusions are consistent with a two-plate model of the Grenville, the evidence is far from conclusive. At present we can only conclude that none of the models so far proposed to explain Grenville paleomagnetic data provide at the same time a natural explanation of the Grenville orogeny. A solution to the Grenville problem [Baer et al., 1974] must await clarification of the relationship between isotopic ages and paleopoles, or the discovery of group A poles outside the Grenville Province.

Conclusion

Paleomagnetic and rock magnetic investigations of the Haliburton intrusions have lead to the following conclusions.

1. Three distinct components of magnetization, recording the late Precambrian magnetic field at three different times, are present in the Haliburton intrusions.

2. These A, B, and C magnetizations have distinctive coercivity and blocking temperature spectra, reflecting conditions of temperature, cooling rate, and chemical reequilibration of the magnetic minerals at the time each remanence was 'frozen-in'.

3. The A component is analogous in its direction (pole at 142.5°E, 35.6°S), its normal polarity, and its high coercive force and blocking temperature to group A magnetizations of other studies. The C component (reversed polarity, pole at 167.4°E, 3.0°S) resembles group B poles of other studies. The B component has distributed low blocking temperatures like the superimposed magnetization Me_2 of the Morin anorthosite [Irving et al, 1974a], but its normal polarity and paleopole (172.3°E, 24.5°N) are like those of late Keweenawan rocks.

4. It is unlikely that any of the three components 'sees through' the high-grade regional metamorphism. Likewise, none postdate the Grenville orogeny.

5. The probable age sequence from oldest to youngest is A-C-B, the A component being acquired as a TRM above 500°C, the B component as a viscous partial TRM between about 350°C and room temperature, and the C component as a TRM or TCRM (thermochemical remanent magnetization) at intermediate temperatures. Although all three components can be accounted for by a single slow cooling event, the C and B components may have been set as a result of thermal pulses superimposed on the main cooling process.

6. The sequence A-C-B is consistent with converging Grenville and 'Interior Laurentian' polar wander paths. However, the polar juncture corresponds in time to the resetting of K/Ar ages in the latter stages of cooling (i.e., during the 950-m.y. Grenville orogeny), not the high-grade regional metamorphism.

Acknowledgments. We are indebted to A. J. Baer, W. F. Fahrig, H. C. Halls, E. Irving, H. C. Palmer, L. Pesonen, W. A. Robertson, J. Roy, and

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R. van der Voo for many spirited discussions of the Grenville problem. D. Buchan, P. DeVries, B. Downing, H. C. Halls, J. A. Hanes, and W. F. Schwerdtner were of great assistance in the field. Polished section studies are due to J. A. Hanes. This project has been supported by the Department of Energy, Mines and Resources and the National Research Council of Canada.

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(Received June 18, 1975;
revised December 16, 1975;
accepted February 20, 1976.)

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